In [18]:
```
#Load and Explore the Dataset
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\Administrator\Downloads\Heart Disease UCI.csv")
df
```

Out[18]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tl
0	69	1	0	160	234	1	2	131	0	0.1	1	1	
1	69	0	0	140	239	0	0	151	0	1.8	0	2	
2	66	0	0	150	226	0	0	114	0	2.6	2	0	
3	65	1	0	138	282	1	2	174	0	1.4	1	1	
4	64	1	0	110	211	0	2	144	1	1.8	1	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	
292	40	1	3	152	223	0	0	181	0	0.0	0	0	
293	39	1	3	118	219	0	0	140	0	1.2	1	0	
294	35	1	3	120	198	0	0	130	1	1.6	1	0	
295	35	0	3	138	183	0	0	182	0	1.4	0	0	
296	35	1	3	126	282	0	2	156	1	0.0	0	0	

297 rows × 14 columns

In [17]:
```
df.head(10)
```

Out[17]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	69	1	0	160	234	1	2	131	0	0.1	1	1	0
1	69	0	0	140	239	0	0	151	0	1.8	0	2	0
2	66	0	0	150	226	0	0	114	0	2.6	2	0	0
3	65	1	0	138	282	1	2	174	0	1.4	1	1	0
4	64	1	0	110	211	0	2	144	1	1.8	1	0	0
5	64	1	0	170	227	0	2	155	0	0.6	1	0	2
6	63	1	0	145	233	1	2	150	0	2.3	2	0	1
7	61	1	0	134	234	0	0	145	0	2.6	1	2	0
8	60	0	0	150	240	0	0	171	0	0.9	0	0	0
9	59	1	0	178	270	0	2	145	0	4.2	2	0	2

In [2]:
```
df.describe()
```

Out[2]:

	age	sex	cp	trestbps	chol	fbs	reste
count	297.000000	297.000000	297.000000	297.000000	297.000000	297.000000	297.000
mean	54.542088	0.676768	2.158249	131.693603	247.350168	0.144781	0.996
std	9.049736	0.468500	0.964859	17.762806	51.997583	0.352474	0.994
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000
25%	48.000000	0.000000	2.000000	120.000000	211.000000	0.000000	0.000
50%	56.000000	1.000000	2.000000	130.000000	243.000000	0.000000	1.000
75%	61.000000	1.000000	3.000000	140.000000	276.000000	0.000000	2.000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000

In [5]:

```
df.dtypes
```

Out[5]:

```
age            int64
sex            int64
cp             int64
trestbps       int64
chol           int64
fbs            int64
restecg        int64
thalach        int64
exang          int64
oldpeak      float64
slope          int64
ca             int64
thal           int64
condition      int64
dtype: object
```

In [183]:

```
df.isnull().sum()
```

Out[183]:

```
age              0
sex              0
cp               0
trestbps         0
chol             0
fbs              0
restecg          0
thalach          0
exang            0
oldpeak          0
slope            0
ca               0
thal             0
condition        0
risk_score       0
risk_category    0
dtype: int64
```

In [8]:

```
print(df.duplicated().sum())
```

```
0
```

In [9]:
```
df.info()
```

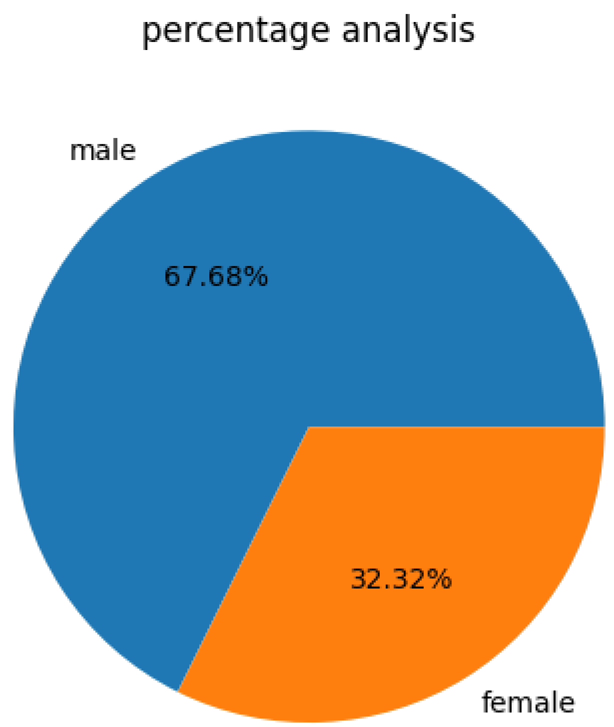
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 297 entries, 0 to 296
Data columns (total 14 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   age        297 non-null    int64  
 1   sex        297 non-null    int64  
 2   cp         297 non-null    int64  
 3   trestbps   297 non-null    int64  
 4   chol       297 non-null    int64  
 5   fbs        297 non-null    int64  
 6   restecg    297 non-null    int64  
 7   thalach    297 non-null    int64  
 8   exang      297 non-null    int64  
 9   oldpeak    297 non-null    float64
 10  slope      297 non-null    int64  
 11  ca         297 non-null    int64  
 12  thal       297 non-null    int64  
 13  condition  297 non-null    int64  
dtypes: float64(1), int64(13)
memory usage: 32.6 KB
```

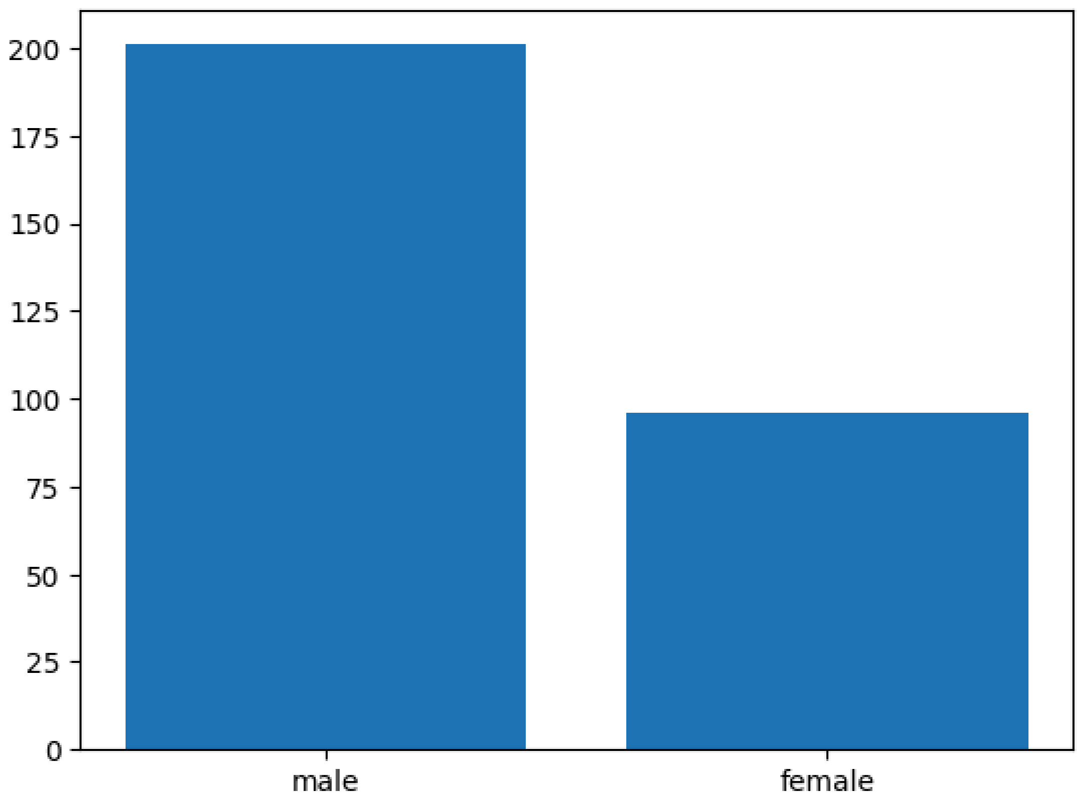
In [10]:
```
df.columns
```

Out[10]:
```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'condition'],
      dtype='object')
```

In [50]:
```
#Gender Distribution Analysis
d=df['sex'].value_counts()
k=["male","female"]
total=len(df)
male=d[1]
male_percent=male/total*100
print("percentage of males are",male_percent)
female=d[0]
female_percent=female/total*100
print("percentage of males are",female_percent)
plt.pie(d,labels=k,autopct='%2.2f%%')
plt.title("percentage analysis")
plt.show()

plt.bar(k,d.values)
plt.show()
```

```
percentage of males are 67.67676767676768
percentage of males are 32.323232323232325
```

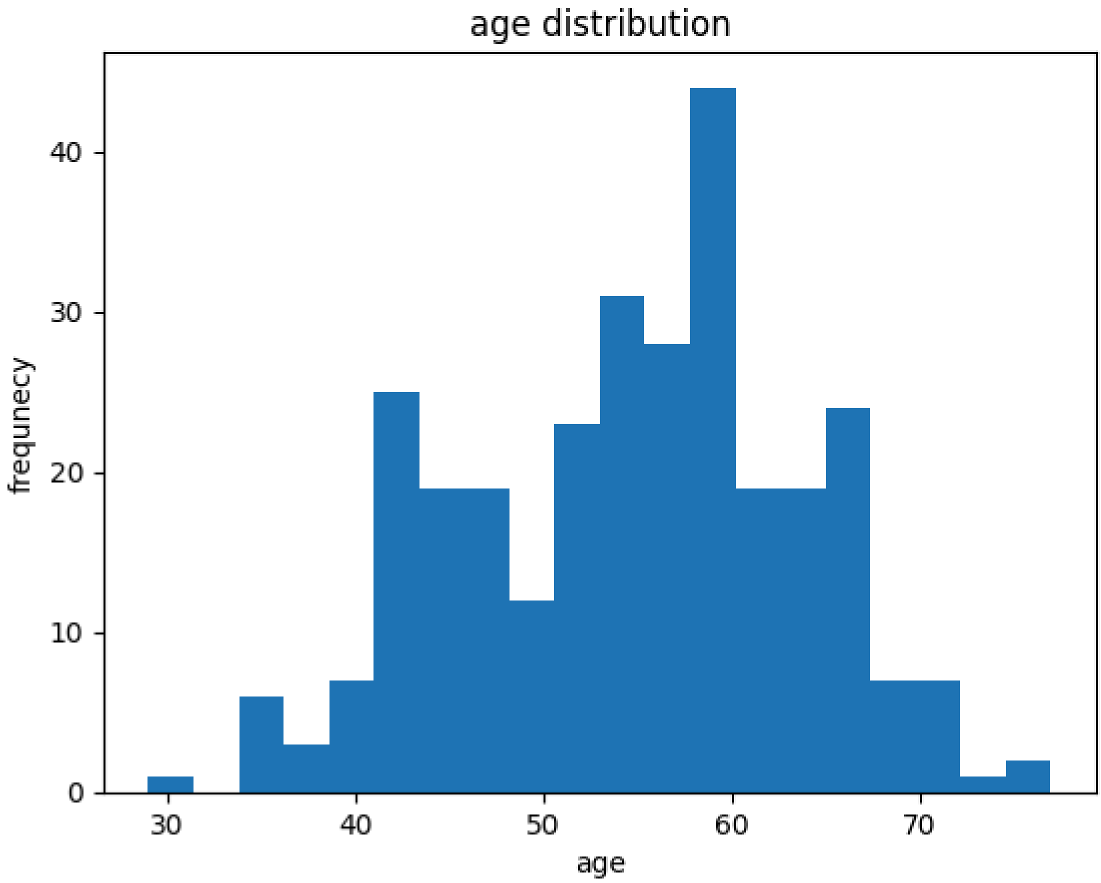




In [79]:
```
#Age Analysis
age=df['age']
print("maximum age is",age.max())
print("minimum age is",age.min())
print("mean age is",age.mean())
print("median age is",age.median())
```

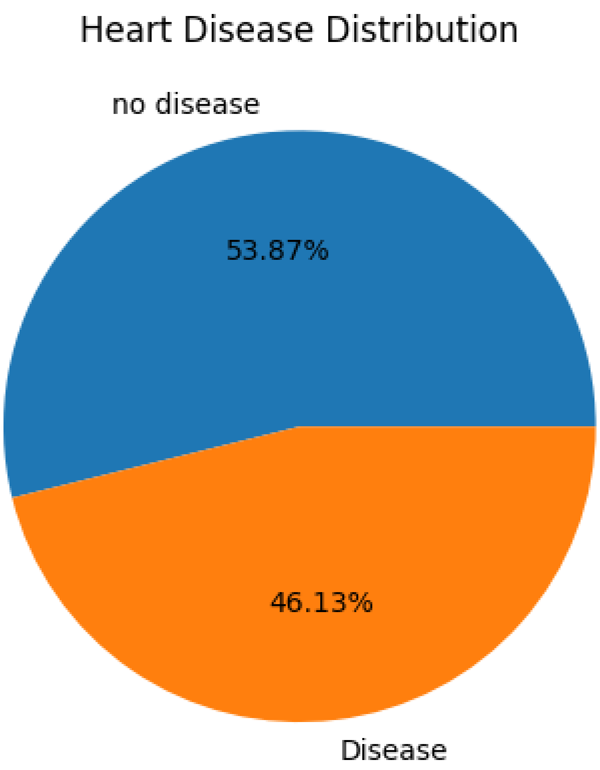
```
maximum age is 77
minimum age is 29
mean age is 54.54208754208754
median age is 56.0
```

In [55]:
```
plt.hist(df['age'],bins=20)
plt.title("age distribution")
plt.xlabel("age")
plt.ylabel("frequnecy")
plt.show()
```



In [168]:
```
# Target Variable Analysis
target_counts = df['condition'].value_counts()
print(target_counts)
plt.figure()
plt.pie(target_counts, labels=["no disease","Disease"], autopct="%2.2f%%")
plt.title("Heart Disease Distribution")
plt.show()

disease_percent = (target_counts[1]/len(df))*100
print("Disease %:", disease_percent)
```

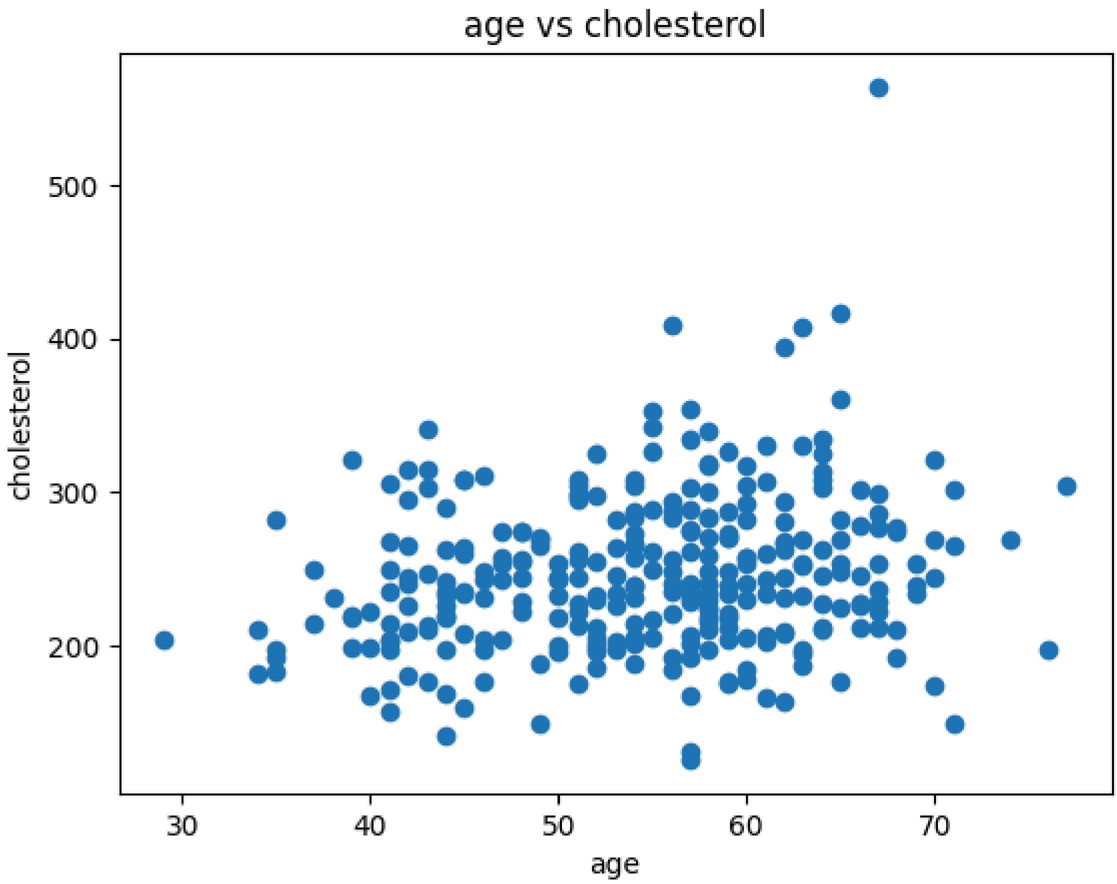
```
condition
0    160
1    137
Name: count, dtype: int64
```




```
Disease %: 46.12794612794613
```

In [152]:
```
#Correlation Between Age and Cholesterol
corr = df[['age','chol']].corr()
print(corr)

plt.figure()
plt.scatter(df['age'], df['chol'])
plt.title("age vs cholesterol")
plt.xlabel("age")
plt.ylabel("cholesterol")
plt.show()
```

```
           age      chol
age   1.000000  0.202644
chol  0.202644  1.000000
```

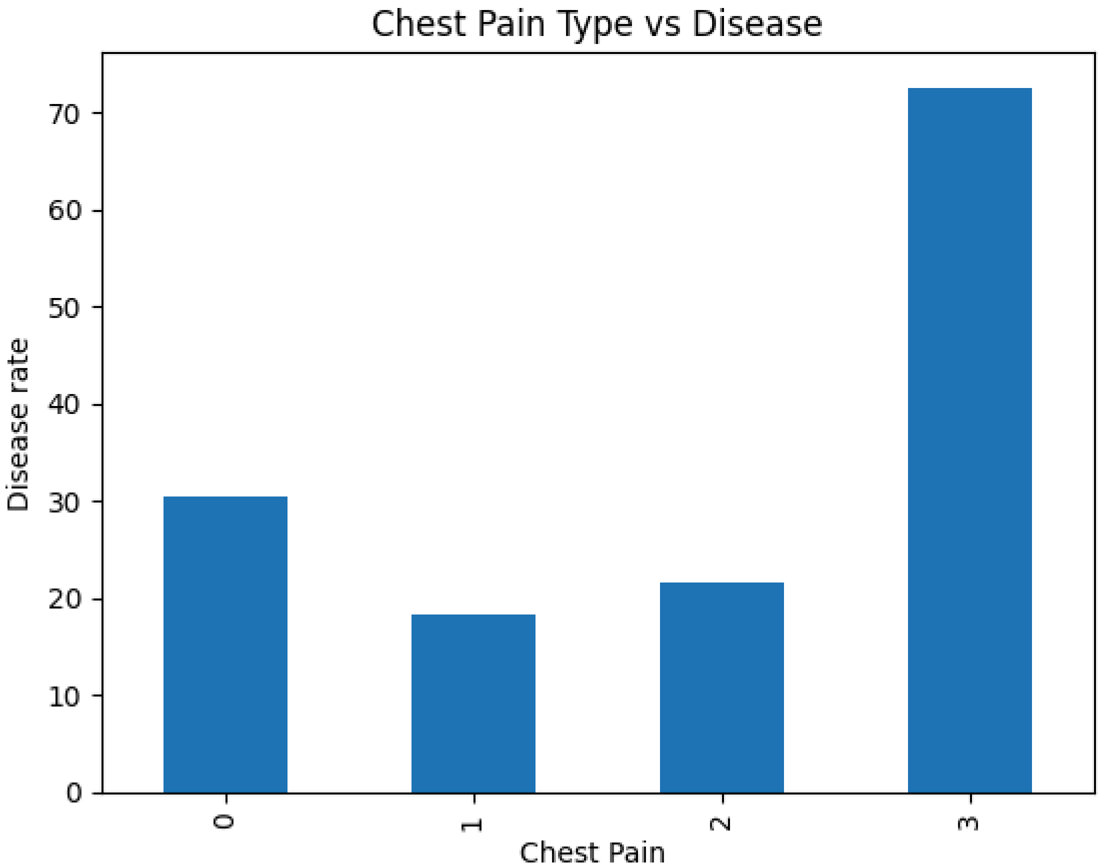


In [177]:
```
# Interpret the relationship between Age and Cholesterol
age_chol_corr = df['age'].corr(df['chol'])
print(f"Correlation between Age and Cholesterol: {age_chol_corr:.2f}")
if abs(age_chol_corr) < 0.3:
    print("Interpretation: There is a weak relationship between age and cholestero
elif abs(age_chol_corr) < 0.7:
    print("Interpretation: There is a moderate relationship between age and choles
else:
    print("Interpretation: There is a strong relationship between age and choleste
if age_chol_corr > 0:
    print("The relationship is positive, means as age increases, cholesterol tends
else:
    print("The relationship is negative, means as age increases, cholesterol tends
```

```
Correlation between Age and Cholesterol: 0.20
Interpretation: There is a weak relationship between age and cholesterol levels.
The relationship is positive, means as age increases, cholesterol tends to increas
e.
```

In [175]:
```
#Chest Pain Type vs Disease
cp = df.groupby('cp')['condition'].mean()*100
print(cp)

cp.plot(kind='bar')
plt.title("Chest Pain Type vs Disease")
plt.xlabel("Chest Pain ")
plt.ylabel("Disease rate")
plt.show()
print("chest pain type which has high risk is",cp.idxmax())
```

```
cp
0    30.434783
1    18.367347
2    21.686747
3    72.535211
Name: condition, dtype: float64
```




```
chest pain type which has high risk is 3
```

In [178]:

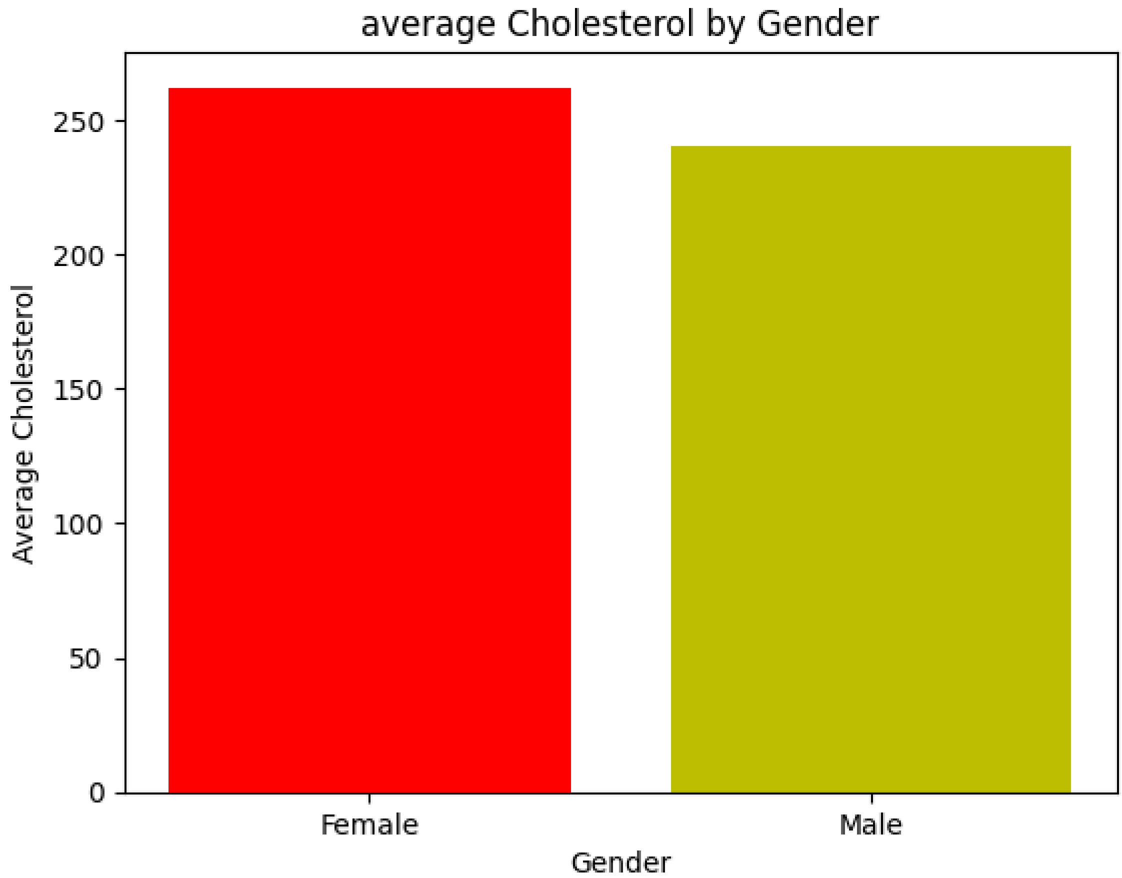
```python
# Identify which chest pain type is most risky
most_risky_cp = cp_group.idxmax()
highest_rate = cp_group.max()
print(f"The chest pain type with the highest disease rate is {most_risky_cp} with
```

```
The chest pain type with the highest disease rate is 3 with a rate of 0.73%.
```

In [167]:

```python
#Average Cholesterol by Gender

sex=df.groupby('sex')['chol'].size()
print(sex)

chol_gender = df.groupby('sex')['chol'].mean()
print(chol_gender)
gender=['Female','Male']
plt.figure()
plt.bar(gender, chol_gender,color=['r','y'])
plt.title("average Cholesterol by Gender")
plt.xlabel("Gender")
plt.ylabel("Average Cholesterol")
plt.show()
```

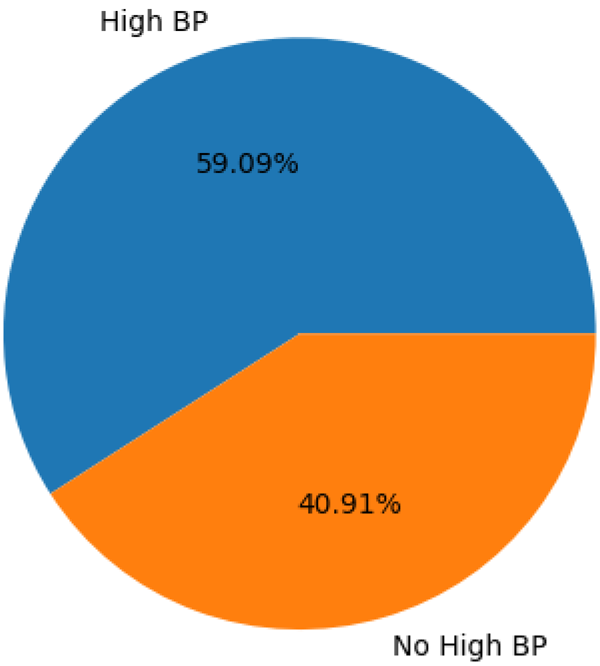
```
sex
0     96
1    201
Name: chol, dtype: int64
sex
0    262.229167
1    240.243781
Name: chol, dtype: float64
```



In [122]:

```
#Resting Blood Pressure Analysis

average_bp = df['trestbps'].mean()
print("Average BP:",average_bp)

hi_bp = df[df['trestbps'] > 140]
print("High BP Patients:", len(hi_bp))

per_disease=high_bp['condition'].mean()*100
print("disease % in High BP:",per_disease)
per_no_disease = 100 - per_disease
values = [per_disease, per_no_disease]
labels = ['High BP', 'No High BP']
plt.pie(values, labels=labels, autopct='%2.2f%%')
plt.show()
```

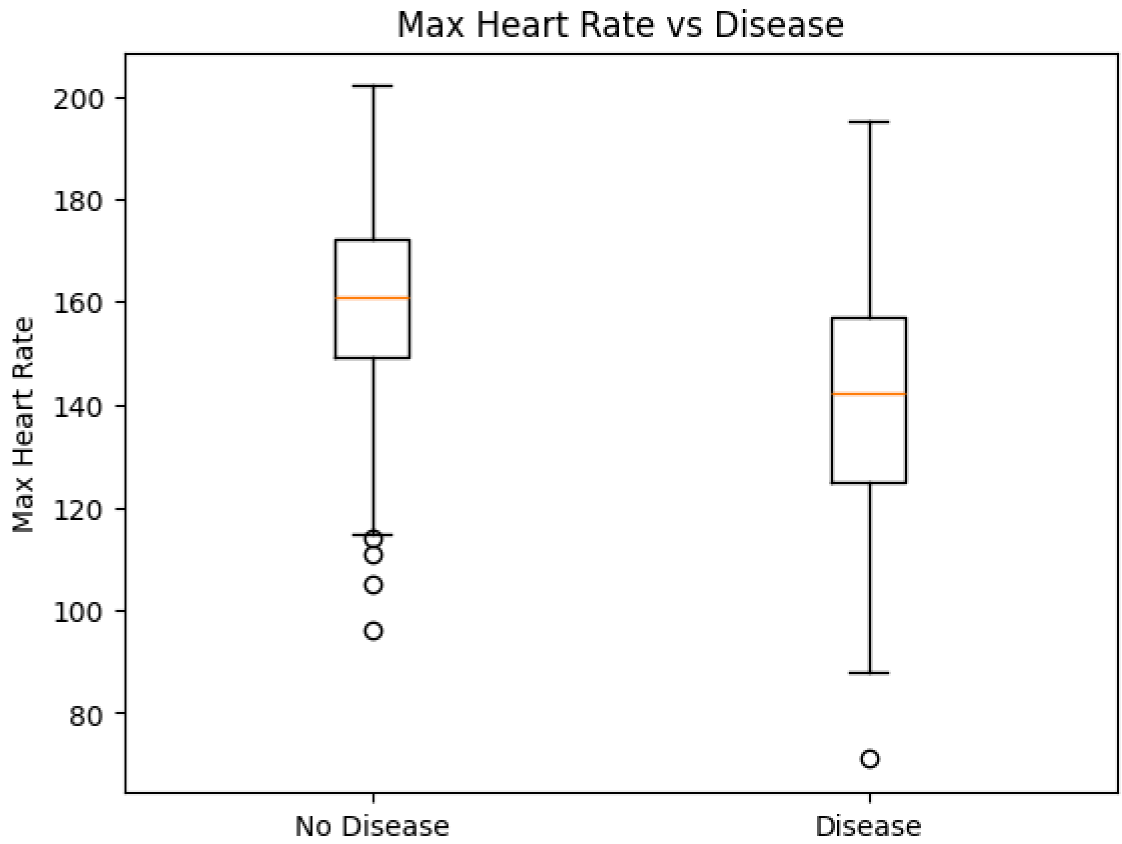
```
Average BP: 131.69360269360268
High BP Patients: 66
disease % in High BP: 59.09090909090909
```



In [176]:

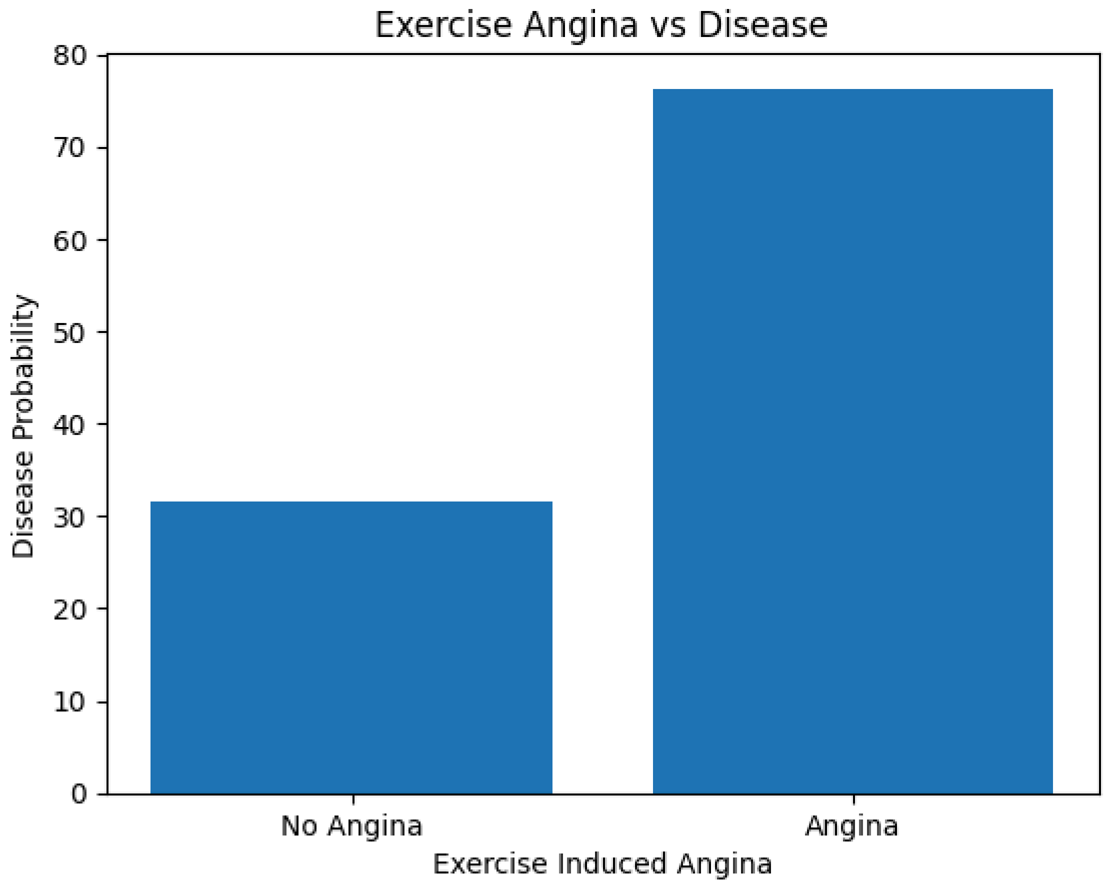
```
#Maximum Heart Rate vs Disease
disease = df[df['condition']==1]['thalach']
no_disease = df[df['condition']==0]['thalach']

print("avg hr disease:", disease.mean())
print("avg hr no disease:", no_disease.mean())

plt.figure()
plt.boxplot([no_disease, disease])
plt.xticks([1,2], ["No Disease","Disease"])
plt.title("Max Heart Rate vs Disease")
plt.ylabel("Max Heart Rate")
plt.show()
```

```
avg hr disease: 139.1094890510949
avg hr no disease: 158.58125
```



In [133]:
```
#Exercise Induced Angina Impact
exang_group = df.groupby('exang')['condition'].mean()
print("percentage with exang",exang_group[1]*100)
print("percentage without exang",exang_group[0]*100)
exang_per=[exang_group[0]*100,exang_group[1]*100]

plt.figure()
plt.bar(['No Angina','Angina'],exang_per)
plt.title("Exercise Angina vs Disease")
plt.xlabel("Exercise Induced Angina")
plt.ylabel("Disease Probability")
plt.show()
```

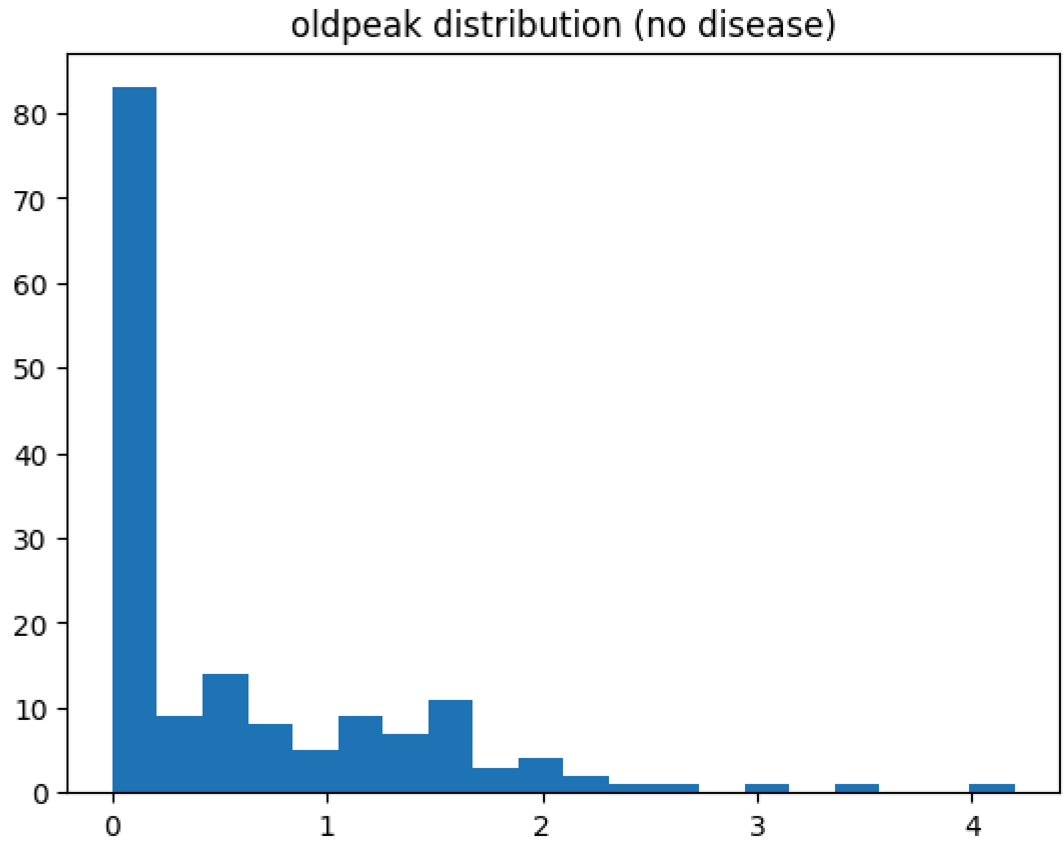
```
percentage with exang 76.28865979381443
percentage without exang 31.5
```

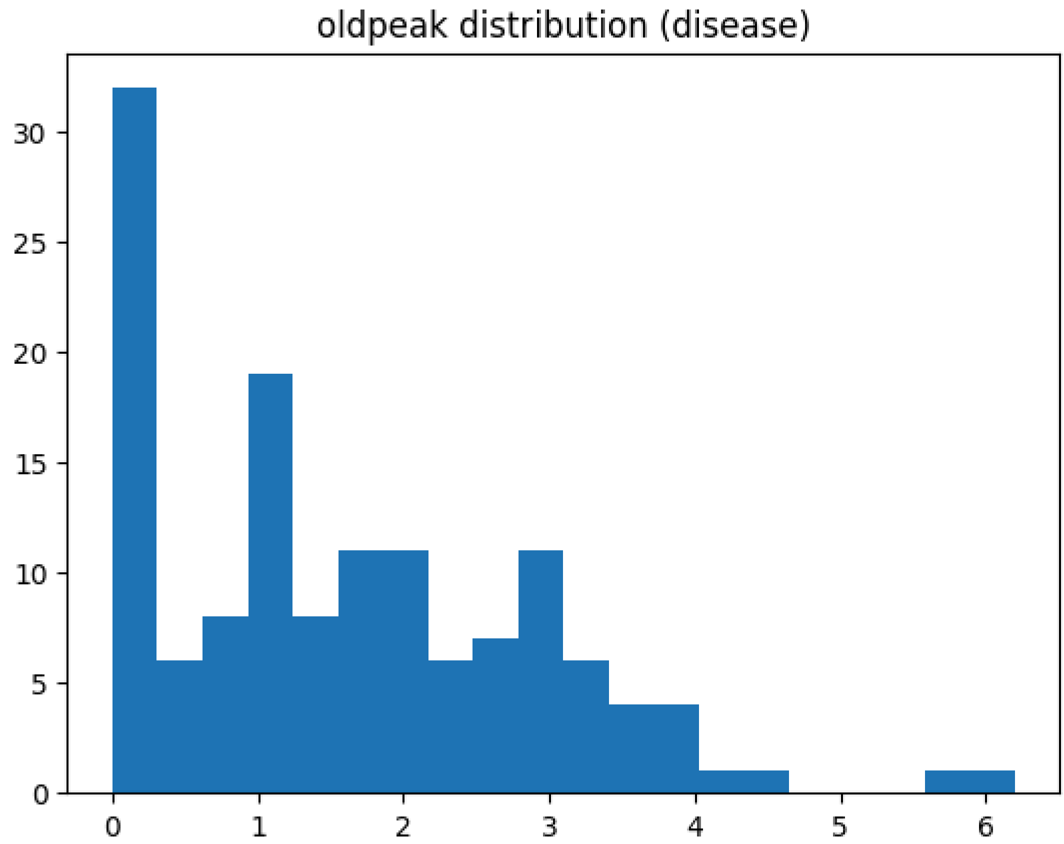


In [180]:

```
#ST Depression (oldpeak) Analysis
oldpeak_group = df.groupby('condition')['oldpeak'].mean()
print(oldpeak_group)

plt.figure()
plt.hist(df[df['condition']==0]['oldpeak'], bins=20)
plt.title("oldpeak distribution (no disease)")
plt.show()

plt.figure()
plt.hist(df[df['condition']==1]['oldpeak'], bins=20)
plt.title("oldpeak distribution (disease)")
plt.show()
```

```
condition
0    0.598750
1    1.589051
Name: oldpeak, dtype: float64
```





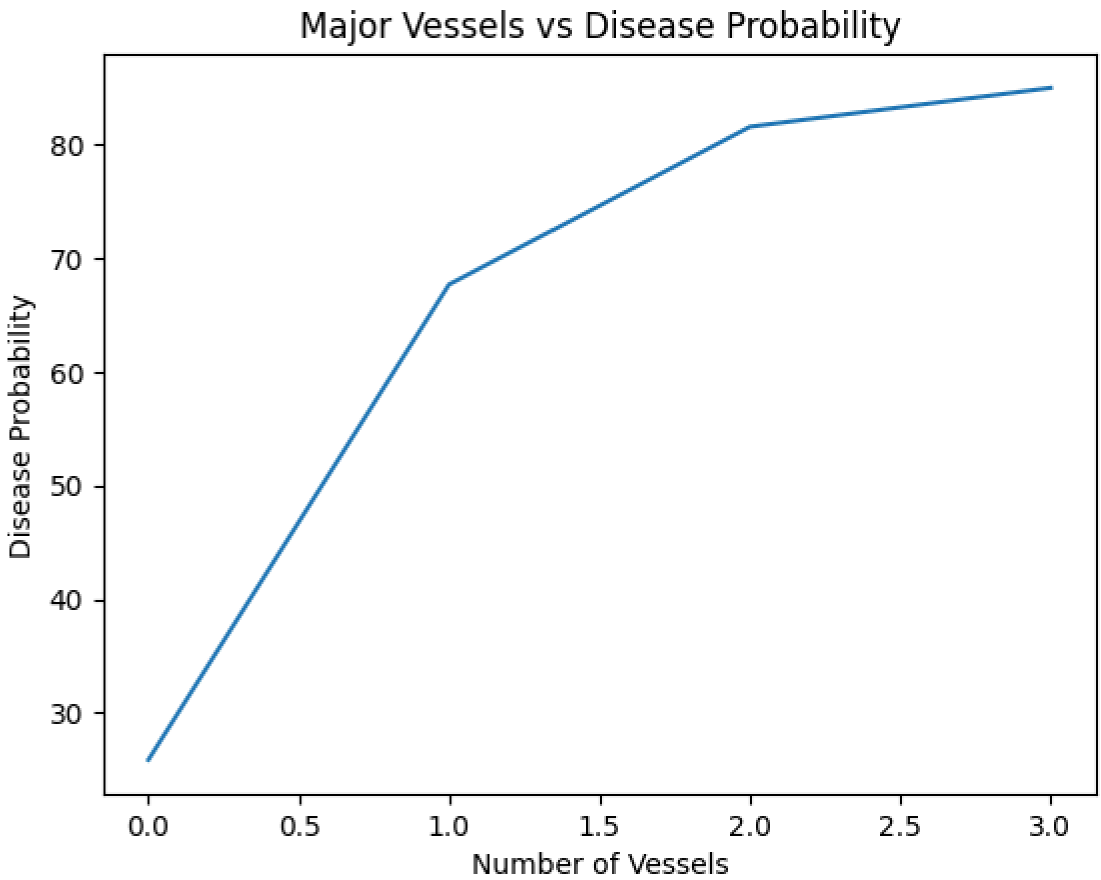
In [181]:

```
# Identify trend
if oldpeak_group[1] > oldpeak_group[0]:
    print("Trend: Patients with heart disease tend to have higher ST depression (c
else:
    print("Trend: Patients with heart disease tend to have lower ST depression (ol
```

```
Trend: Patients with heart disease tend to have higher ST depression (oldpeak)
```

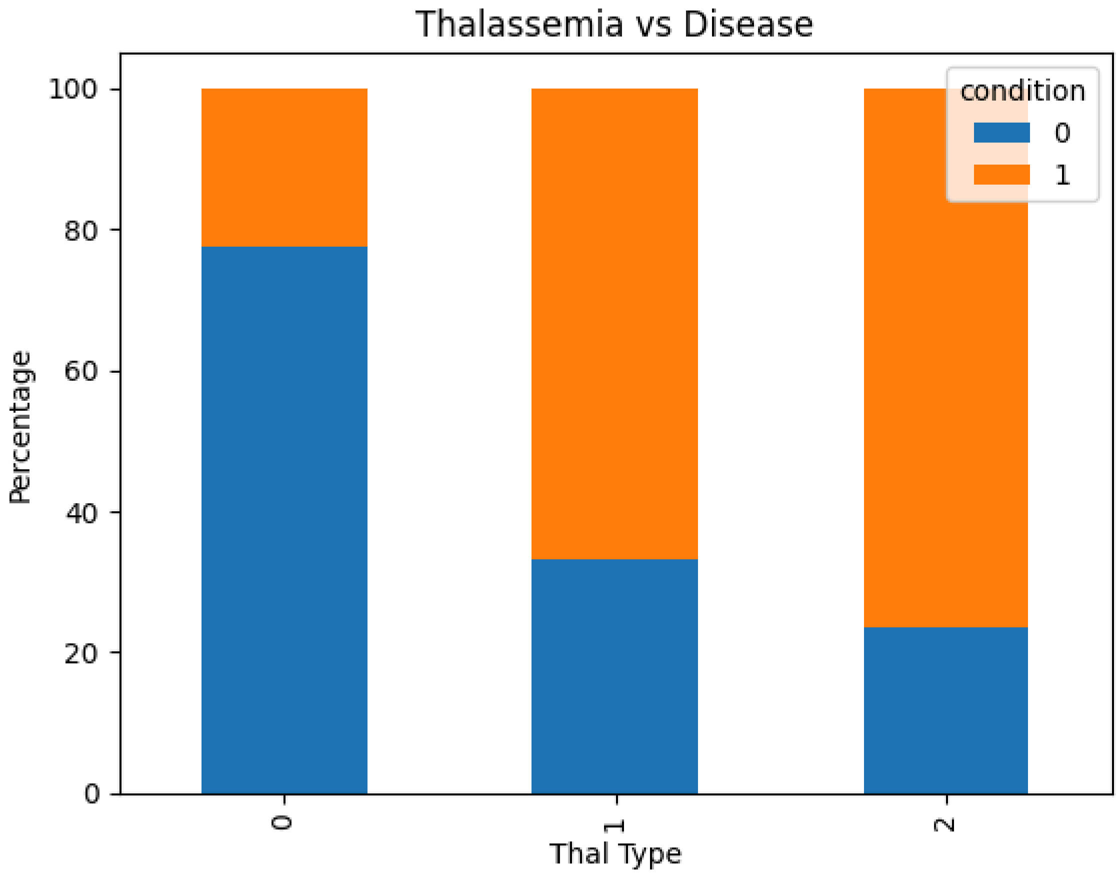
In [169]:
```
#Number of Major Vessels (ca) Impact
ca_group = df.groupby('ca')['condition'].mean()*100
print(ca_group)

plt.figure()
plt.plot(ca_group.index, ca_group.values)
plt.title("Major Vessels vs Disease Probability")
plt.xlabel("Number of Vessels")
plt.ylabel("Disease Probability")
plt.show()
```

```
ca
0    25.862069
1    67.692308
2    81.578947
3    85.000000
Name: condition, dtype: float64
```



In [161]:
```
#Thalassemia vs Disease
#thal_ct = pd.crosstab(df['thal'], df['condition'], normalize='index')
#print(thal_ct)

thal_ct = pd.crosstab(df['thal'], df['condition'], normalize='index')*100
print(thal_ct)

thal_ct.plot(kind='bar', stacked=True)
plt.title("Thalassemia vs Disease")
plt.xlabel("Thal Type")
plt.ylabel("Percentage")
plt.show()
```

```
condition          0          1
thal
0          77.439024  22.560976
1          33.333333  66.666667
2          23.478261  76.521739
```

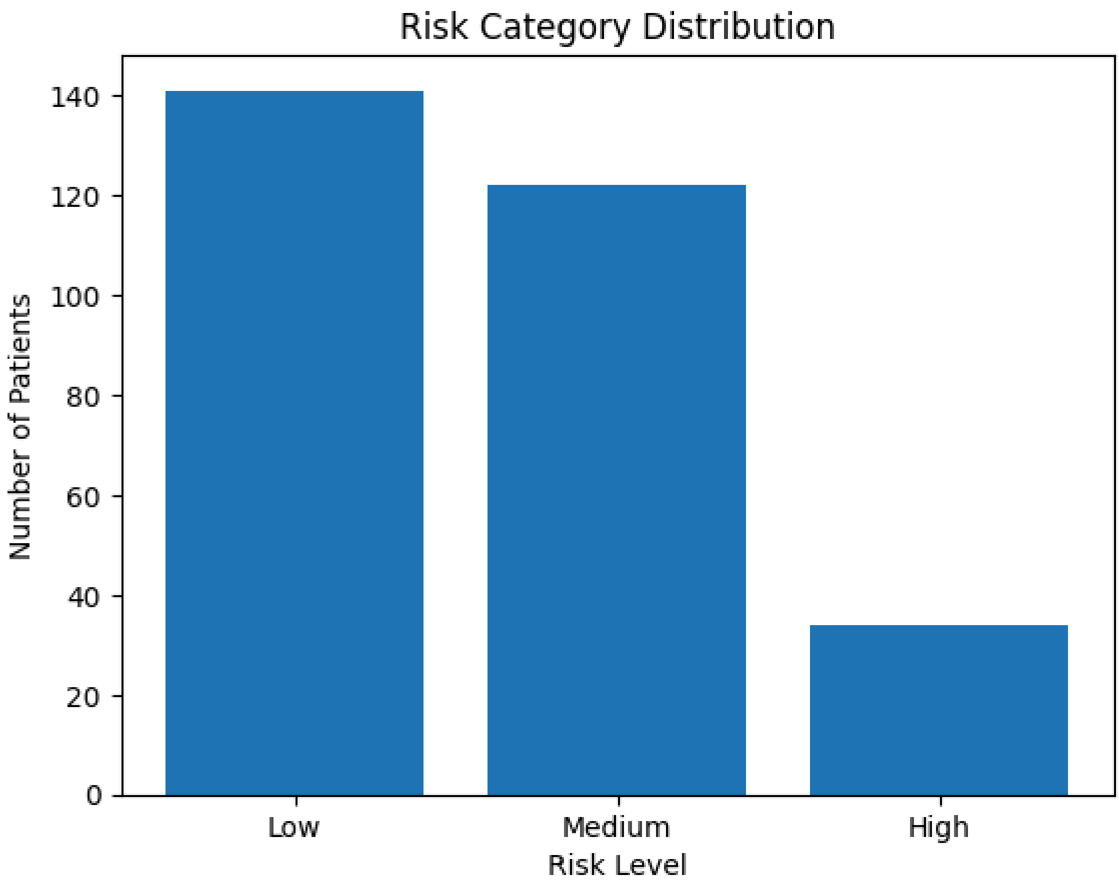


In [158]:
```python
#Multi-Factor Risk Analysis
high_risk = df[(df['age'] > 50) &(df['chol'] > 240) &(df['trestbps'] > 140)]

print("Total High Risk Patients:", len(high_risk))

print("Disease % in High Risk:",
      high_risk['condition'].mean()*100)
```

```
Total High Risk Patients: 33
Disease % in High Risk: 66.66666666666666
```

In [160]:
```python
#Create Risk Score (Custom Analysis)
df['risk_score'] = (df['chol']/200) + (df['trestbps']/120) + df['oldpeak']

def classify(score):
    if score < 3:
        return "Low"
    elif score < 5:
        return "Medium"
    else:
        return "High"

df['risk_category'] = df['risk_score'].apply(classify)

risk_dist = df['risk_category'].value_counts()

plt.figure()
plt.bar(risk_dist.index, risk_dist.values)
```

```
plt.title("Risk Category Distribution")
plt.xlabel("Risk Level")
plt.ylabel("Number of Patients")
plt.show()
```



In [182]:

```
df['risk_score'] = (df['chol'] / 200) + (df['trestbps'] / 120) + df['oldpeak']

# Classify patients as Low Risk, Medium Risk, High Risk
def classify_risk(score):
    if score < 3:
        return 'Low Risk'
    elif score <= 5:
        return 'Medium Risk'
    else:
        return 'High Risk'

df['risk_category'] = df['risk_score'].apply(classify_risk)

print("Risk score statistics:")
print(df['risk_score'].describe())
print("\nRisk category distribution:")
print(df['risk_category'].value_counts())
```

```
Risk score statistics:
count    297.000000
mean       3.389753
std        1.244547
min        1.733333
25%        2.393333
50%        3.075000
75%        4.091667
max        8.353333
Name: risk_score, dtype: float64

Risk category distribution:
risk_category
Low Risk       141
Medium Risk    122
High Risk       34
Name: count, dtype: int64
```

In [ ]:

```
1. Does cholesterol strongly impact heart disease?
  Yes high cholesterol is a major risk factor for heart disease.
Managing cholesterol through diet, exercise, and medical care is crucial for preve

2. Is the male population more vulnerable to heart disease?
   compared to women ,men has the higher chances of heartattack.womens risk increa

3. Does exercise-induced angina significantly increase risk?
    Yes, exercise-induced angina is a warning sign of underlying coronary artery c
    It shows that the oxygenlevel to the heart is not  upto level physical exertic
    which significantly raises the risk of serious cardiac events if left untreate
4. Which feature has the strongest correlation with heart disease?
   from these features, exercise-induced angina shows the strongest correlation wi
    While age, cholesterol, and gender are important, angina directly reflects cor
    making it a powerful predictor of disease presence.
```

In [ ]: